

ICT80004 Weekly Communication – Week #05

Student Name: Arun Ragavendhar Arunachalam Palaniyappan ID: 104837257

Organisation: Commonwealth Scientific and Industrial Research Organisation (CSIRO)

Industry Supervisor: Dr. Shigang Liu

Date Prepared: 05/09/2025 Internship Week #: 5

Day	Date	Task(s) completed
1	Monday 1 Sep 2025 8 hours	- Finalised the code-based testing suite by adding seven new cases: use-after-free, double free, buffer overflow, format string, command injection, path traversal, and uninitialised variable use. - Completed tables for all models (GPT-4/5, Grok, Gemini, Llama-3, DeepSeek V1) across four prompting styles (Chain of Thought, Chain of Action, Chain of Debate, Encouragement Learning).
2	Wednesday 3 Sep 2025 8 hours	- Verified fixes by compiling/running where applicable and recorded outcomes (compiles/runs, vulnerability prevented, notes). - Consolidated all code-based results (Weeks 3–5) into one document with explanations and inferences.

Total hours completed for the week: 16

Plans for next week: #06 week (8 - 12 Sept 2025)

Start integrating boundary/negative inputs into the test suite (e.g., LLONG_MAX, invalid indices, crafted SQL payloads).

Map each code test to the taxonomy and begin drafting the **Prompt Injection Test Suite** structure for CSIRO platform models.

Prepare a brief slide deck to present the completed code-based phase and the next steps.

Screenshot of Timely EMAIL communication update to the Supervisor at the end of week #05 sent on 05 September 2025 Friday at 3:16 PM (15:16).

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Weekly Communication / Reflection update - #05 week, 1 - 5 Sep 2025

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Arun Ragavendhar <arunragavendhar.1999@gmail.com>

15:16 (1 minute ago)

to Shigang

Dear Dr. Shigang Liu,

I hope you are well. Please find below my update for Week 5 of the internship.

Activities completed this week (Total: 16 hours):

- Finalised the code-based testing suite by adding seven new cases: use-after-free, double free, buffer overflow, format string, command injection, path traversal, and uninitialised variable use.
- Completed tables for all models (GPT-4/5, Grok, Gemini, Llama-3, DeepSeek V1) across four prompting styles (Chain of Thought, Chain of Action, Chain of Debate, Encouragement Learning).
- Verified fixes by compiling/running where applicable and recorded outcomes (compiles/runs, vulnerability prevented, notes).
- Consolidated all code-based results (Weeks 3–5) into one document with explanations and inferences.

Plan for next week (8–12 Sept 2025):

- Start integrating boundary/negative inputs into the test suite (e.g., LLONG_MAX, invalid indices, crafted SQL payloads).
- Map each code test to the taxonomy and begin drafting the **Prompt Injection Test Suite** structure for CSIRO platform models.
- Prepare a brief slide deck to present the completed code-based phase and the next steps.

Kind regards,
Arun Ragavendhar Arunachalam Palaniyappan
ICT80004 Internship Student – CSIRO Data61

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